

Mayank Goud Polagoni

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Education

Northeastern University, Boston, MA

May 2026

*Master of Science in Robotics, **Concentration** : Electrical and Computer Engineering*

Relevant Coursework: Robotic Sensing and Navigation, Mobile Robotics, Control Systems Engineering, Robot Mechanics and Control, Reinforcement Learning, Pattern Recognition and Computer Vision

CVR College Of Engineering, Hyderabad, India

2019 – 2023

Bachelor of Technology in Electrical and Electronics Engineering

Technical Skills

Programming & Software: Python, C++, MATLAB, ROS / ROS 2, Linux, Git, Docker, Gazebo, LaTeX, Real-Time Pipelines

Computer Vision & Perception: OpenCV (C++ / Python), YOLOv8, MediaPipe, SIFT / ORB / Harris Corners, SLAM (RTAB-Map), Loop Closure, Bag-of-Words, RGB-D Processing, Multi-Object Tracking, Camera Calibration & AR

Machine Learning & Reinforcement Learning: PyTorch, NumPy, SciPy, Scikit-learn, Deep RL (PPO, SAC, TD3, DDPG, REINFORCE), Policy Gradient, Actor-Critic, OpenAI Gymnasium, CNN, Vision Transformers

Estimation, Control & Sensors: Kalman / Extended Kalman Filter, Hungarian Algorithm, MPPI, Gaussian Process Planners, PID, State-Space Control, Lyapunov, LiDAR, IMU (Allan Variance), RealSense D435i, Jetson Orin NX

Projects

Deep Reinforcement Learning for Autonomous Racing in F1TENTH

Jan 2026 – Apr 2026

Northeastern University

- Implemented PPO, SAC, and TD3 deep RL algorithms in PyTorch for autonomous racing in the F1TENTH Gym simulator using continuous LiDAR-based observations and Ackermann steering dynamics
- Designed reward functions and Gymnasium-compatible environment wrappers, conducting systematic comparison against classical baselines (Pure Pursuit, reactive LiDAR controllers) across five tracks
- Achieved 2× the reward of the best classical baseline with SAC (avg reward 0.289, avg speed 2.89 m/s) and a peak speed of 3.0 m/s with TD3, showing clear off-policy superiority over PPO
- Conducted multi-seed stability analysis and reward-shaping ablations, evaluating generalization to an unseen Berlin track and reporting findings in an AAAI-format report

Real-Time Multi-Object Tracking with Custom Data Association

Jan 2026 – Apr 2026

Northeastern University

- Built a real-time multi-object tracking pipeline from scratch, combining YOLOv8n detection with a custom Kalman Filter for motion prediction and the Hungarian algorithm for data association
- Implemented two cost matrices (IoU and Euclidean distance) for assignment, enabling empirical comparison of matching strategies under occlusion and fast-motion scenarios
- Achieved real-time throughput of 5.9–8.9 FPS on a consumer laptop while maintaining stable identity assignment across frames, with graceful handling of partial occlusion

RTAB-Map Onboard COBRA

Sep 2024 – Dec 2024

Northeastern University

- Implemented RTAB-Map SLAM on COBRA (Crater Observing Bio-inspired Rolling Articulator) robot using RGB-D cameras and Jetson Orin NX for real-time mapping
- Optimized SLAM performance by benchmarking odometry against OptiTrack ground truth, enhancing system accuracy for challenging sidwinding gaits
- Analyzed and addressed sparse point cloud issues in low-object environments by evaluating RTAB-Map's limitations and proposing LiDAR integration for future enhancements
- Collaborated within an interdisciplinary team to collect and process experimental data in varying environments, ensuring robust evaluation of mapping algorithms

Work Experience

Defence Research and Development Laboratory (DRDL)

Oct 2022 – Jan 2023

Electrical Engineering Intern

Hyderabad, India

- Developed a MATLAB state feedback controller, utilizing analytical thinking to conduct parameter analysis and enhance control accuracy
- Applied pragmatic problem-solving to implement advanced control strategies, improving actuator efficiency and dynamic response
- Implemented advanced control strategies like pole placement and adaptive tuning to improve actuator efficiency, reduce energy consumption, and enhance dynamic response
- Refined closed-loop control algorithms to minimize oscillations, achieving a 35% stability improvement